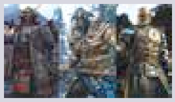
Apple tax evasion
Apple evaded tax payments in New Zealand in spite of bringing in $4.2 billion in sales
http://dgit.in/ApplTax



1,500 banned 'For Honor'
Ubisoft issued a three-day ban to 1,500 players for AFK farming and further gave warnings to 4,000+ players. **http://dgit.in/banForHonor**

HOW WE TESTED

Testing audio gear – and especially headphones – is a minefield to navigate. Audio is so subjective that the old adage of one man's food being another man's poison actually holds true here. Personal preferences overwhelmingly dictate what is considered good or bad. To give you an example of how this does not apply to other product categories we test, consider the case of LCD displays. If a particular display happens to have a slightly warm hue, such a deviation from a neutral colour temperature would be thought of as an inaccuracy and even penalised – no matter how 'pleasing' that inaccuracy would appear to the reviewer. Doing so in audio raises innumerable heckles.

This phenomenon of whatever is 'pleasing' getting better reviews, has pushed some audio manufacturers to tune their gear to produce those particular sound signatures. But the fact remains that such headphones are technically 'inaccurate' and might not work well with certain genres of music or certain mixes. While testing audio, reviewer bias or subjectivity comes in as a result of preferring say bass over treble (in the most simplistic form). Though in an ideal scenario, headphones should be judged by how "true-to-source" their audio output actually is. That is, does the headphone reproduce a particular audio track the way the sound engineer mixing it intended it to sound. How can one know that?

Instruments that output frequency response curves is one way, but the trouble with those is unless the equipment is top notch, such tests produce very little variance for it to be meaningful at all. In this case a "reference" speaker or headphone helps assign a benchmark against which to map deviations. This reference device should be one that's generally known or proven to be flat response. Flat response is a simple concept to understand: it means that the frequency curve produced by the speaker or headphone is flat (or near flat). That is, no frequency range – lows (bass) or highs (treble) – is favoured over the other.

Minimising subjectivity is the holy grail of audio testing but as you can imagine this is a difficult task because as they say, ultimately the proof of the pudding is in the eating. Currently, listening to an audio device is the only way evaluate it. Thus to put our in-ear headphones through their paces properly, we revamped our test process this time around. A few more test tracks were added to the mix in order to properly represent multiple genres. The tracks had a sampling rate of at least 96 kHz (some even 192 kHz when available) and a bit depth of 24 bits (as opposed to the standard bit depth of 16 bits). A few legacy tracks were still deliberately kept in the mix for reproducing real world scenarios but even those necessarily had a minimum bitrate of 320kbps.

As for track selection we used a series of tracks (some of which we've been using for years now) to test specific attributes such as stereo separation, vocal tonality, detail (when instrument density is high), dynamic range etc. As most of you already know, we have a penchant for guitar driven music, but for this test we've deliberately diversified our genres than ever before. There's Ed Sheeran's latest chartbuster thrown in (gasp!).

Apart from these performance tests the build, comfort and features were rated. Longer, woven cables for instance got a better score. This time around we also scored attributes such as isolation and subjective ones such as soundstaging and aesthetics; although the last two didn't get too much weightage in the overall score.

For testing these IEMs we decided to ditch our trusty old ASUS Xonar Essence STX and go with the trendy, portable new high resolution digital audio player from FiiO – the X5 Gen 3. It's got some of the best internals in the industry today and helps isolate against any random buzz or leak that might creep in because of faulty wiring in the AC supply. It's able to drive headphones with high impedance and lastly it's just convenient to quickly be able to change headphones on the fly.

As for other equipment used in the test, you'll keep finding mentions to a few reference headphones such as the RHA S500i, M6Pro, Bose QC20, and the UE 900. These headphones are used for comparison but not specifically thrown into the scoring mix because they have been tested previously with our older test process comprising of a different source and sample track list. They still however make for a good reference point with which to assess these latest models to enter the audio arena.

Very well, not that you know all about our test process, here's another important question: How should you, as a reader and prospective buyer, interpret these scores? Firstly think of the scores assigned as ordinal numbers rather than cardinal ones. The winner of this test (whatever it may be) is the headphone that's best adept at producing a wide range of genres. However some genres and types of music thrive only when a particular frequency range has been boosted. What we'd suggest is look for the row with your type of music in the table and see which headphone fared better. Beyond that for the undecided buyer we also have a "ready reckoner" chart which simplifies the choices even further.

To evaluate the best buy, we have ignored the MRP, which is often inflated. Instead we have taken the undiscounted street price or typical online selling price to reach our "value for money" justifications.

Lastly a small note on nomenclature: the terms in-ears, IEMs and headphones have been used interchangeably in this test. Purists hold your peace and the rest of you - enjoy.

should definitely read our test process above.

BRAINWAVZ S3

We've been seeing quite a few nice offerings from this Chinese manufacturer of affordable audio gear (eg the Omega). The Brainwavz S3 is from its midrange offerings. From the packaging itself you can see that there is emphasis on quality.

Build & Design: The metal housing on the S3 feels solid, while the flat cable remains reasonably tangle free. In order to avoid cord microphonics (a common problem with flat cables) you get a shirt clip in the box. There's are also a number of bi-flange and tri-flange tip options including a set of Comply tips from its T-Series. The housing may feel a bit large but it isn't heavy, making it feel comfortable in spite of the bulk. In terms of looks, we have to agree the S3 looks sleek – it was one of the better looking IEMs in the test. The tips with their translucent grey design go very well with the overall design language of the headphones. The three button in-line remote is designed to work with Apple iOS devices and there's a clear warning on the box which says volume control might not work with Android. But guess what, they worked just fine on a Galaxy S6 Edge and a Google Pixel. The double tap for forward track and triple tap for rewind didn't work fine on the Pixel though. The buttons are annoyingly hard to press but that might just make them last longer I suppose.